

Lab 1: Packet analysis at application layer using Wireshark
SCSR1213 Network Communications
Universiti Teknologi Malaysia

Objective:

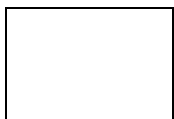
1. Understanding of network protocols by observing the sequence of messages exchanged between two protocol entities, delving down into the details of protocol operation, and causing protocols to perform certain actions and then observing these actions and their consequences.
2. To introduce student with Wireshark software tool for packet analyzer.
3. To analyze protocol used in application layer such as http and dns.

Reference material: Computer Networking: A Top-Down Approach, 7th ed., J.F. Kurose and K.W. Ross.

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Section : **02**



Mark

PART A: Wireshark Getting Started

1.0 Introduction

The basic tool for observing the messages exchanged between executing protocol entities is called a **packet sniffer**. As the name suggests, a packet sniffer captures (“sniffs”) messages being sent/received from/by your computer; it will also typically store and/or display the contents of the various protocol fields in these captured messages. A packet sniffer itself is passive. It observes messages being sent and received by applications and protocols running on your computer, but never sends packets itself. Similarly, received packets are never explicitly addressed to the packet sniffer. Instead, a packet sniffer receives a *copy* of packets that are sent/received from/by application and protocols executing on your machine.

Figure A.1 shows the structure of a packet sniffer. At the right of Figure 1 are the protocols (in this case, Internet protocols) and applications (such as a web browser or ftp client) that normally run on your computer. The packet sniffer, shown within the dashed rectangle in Figure A.1 is an addition to the usual software in your computer, and consists of two parts. The **packet capture library** receives a copy of every link-layer frame that is sent from or received by your computer. In Figure A.1, the assumed physical media is an Ethernet, and so all upper-layer protocols are eventually encapsulated within an Ethernet frame. Capturing all link-layer frames thus gives you all messages sent/received from/by all protocols and applications executing in your computer.

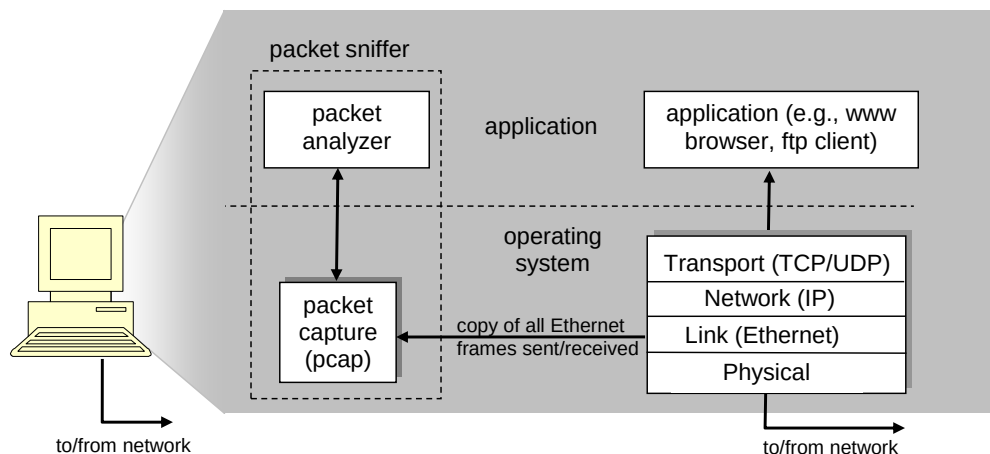


Figure A.1: Packet sniffer structure

The second component of a packet sniffer is the **packet analyzer**, which displays the contents of all fields within a protocol message. In order to do so, the packet analyzer must “understand” the structure of all messages exchanged by protocols. The packet analyzer understands the format of Ethernet frames, and so can identify the IP datagram within an Ethernet frame. It also understands the IP datagram format, so that it can extract the TCP segment within the IP datagram. Finally, it understands the TCP segment structure, so it can extract the HTTP message contained in the TCP segment. Finally, it understands the HTTP protocol and so, for example, knows that the first bytes of an HTTP message will contain the string “GET,” “POST,” or “HEAD”.

2.0 Getting Wireshark Ready

- Download and install the Wireshark software
- Run Wireshark. Wireshark startup screen shown in Figure A.2.

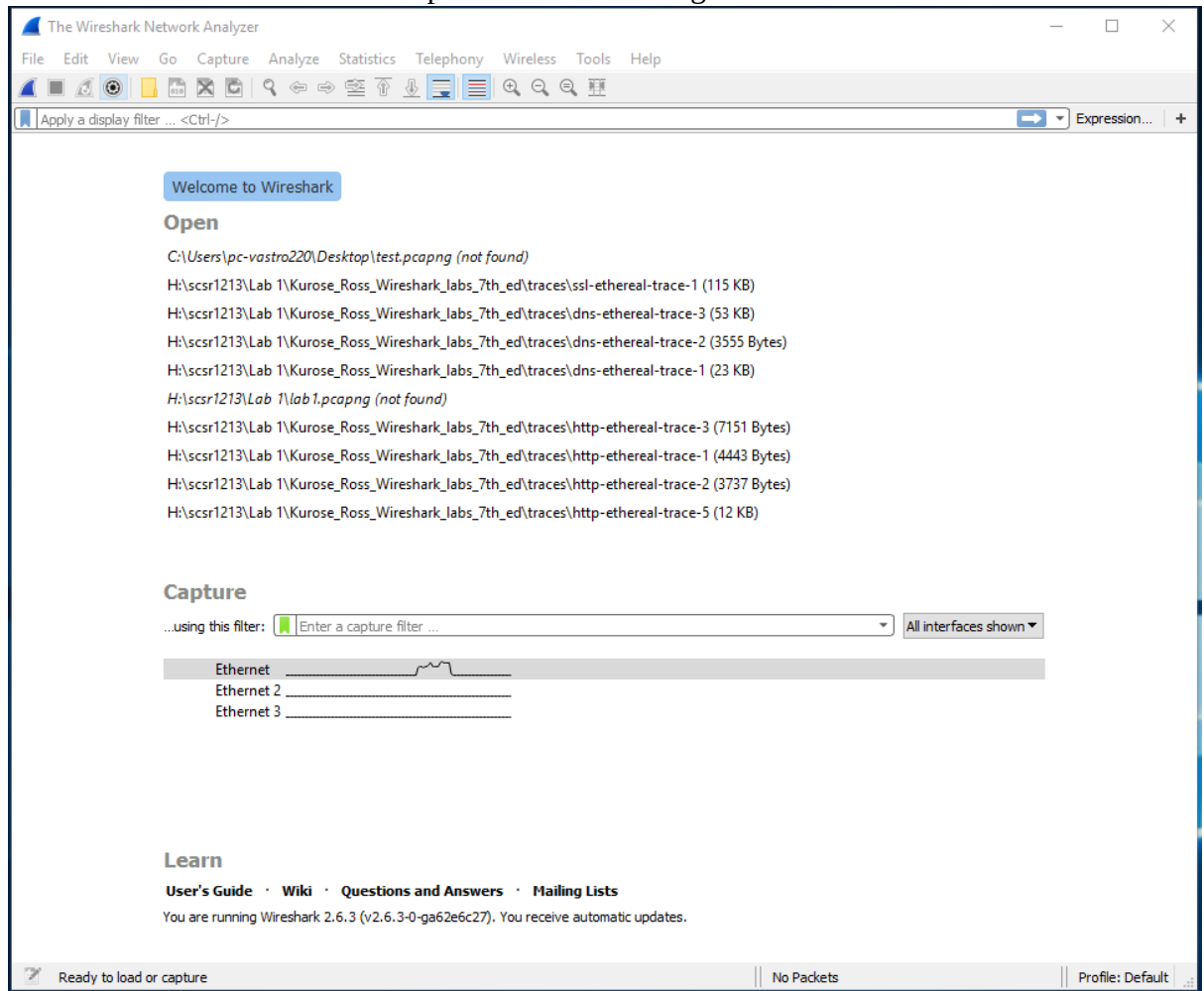


Figure A.2: Initial Wireshark startup screen

- The Wireshark interface has five major components as shown in Figure A.3.

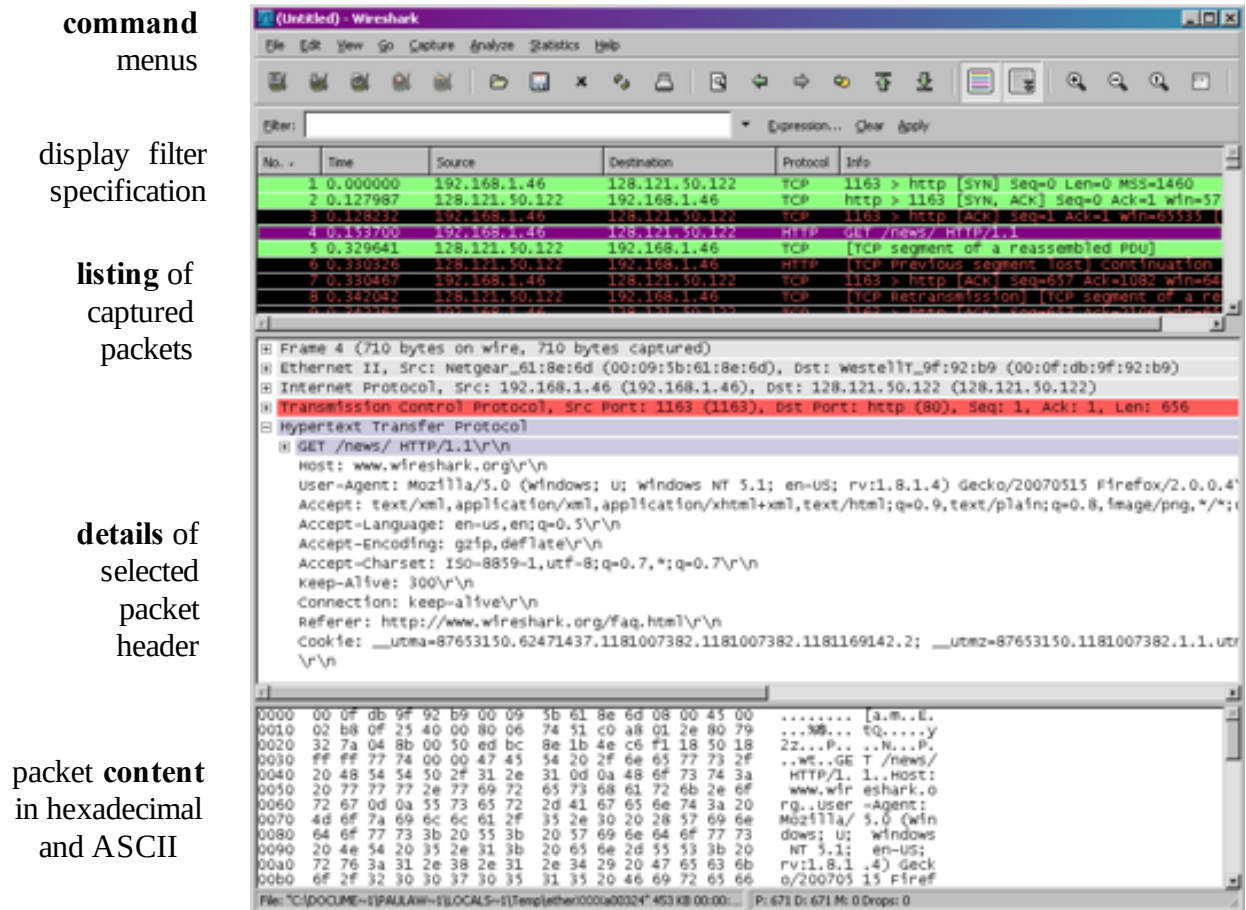


Figure A.3: Wireshark Graphical User Interface, during packet capture and

- The **command menus** are standard pulldown menus located at the top of the window.
- The **packet display filter field**, into which a protocol name or other information can be entered in order to filter the information displayed in the packet-listing window.
- The **packet-listing window** displays a one-line summary for each packet captured, including the packet number, the time at which the packet was captured, the packet's source and destination addresses, the protocol type, and protocol-specific information contained in the packet.
- The **packet-header details window** provides details about the packet selected (highlighted) in the packet-listing window. These details include information about the Ethernet frame and IP datagram that contains this packet. The amount of Ethernet and IP-layer detail displayed can be expanded or minimized by clicking on the plus minus boxes to the left of the Ethernet frame or IP datagram line in the packet details window. If the packet has been carried over TCP or UDP, TCP or UDP details will also be displayed, which can similarly be expanded or minimized. Finally, details about the highest-level protocol that sent or received this packet are also provided.
- The **packet-contents window** displays the entire contents of the captured frame, in both ASCII and hexadecimal format.

3.0 Test Run Wireshark

- Start up the Wireshark software.
- To begin packet capture, select the Capture pull down menu and pick Options menu. Select appropriate interfaces on your compute and click Start button to begin packet capture. Refer to Figure A.4

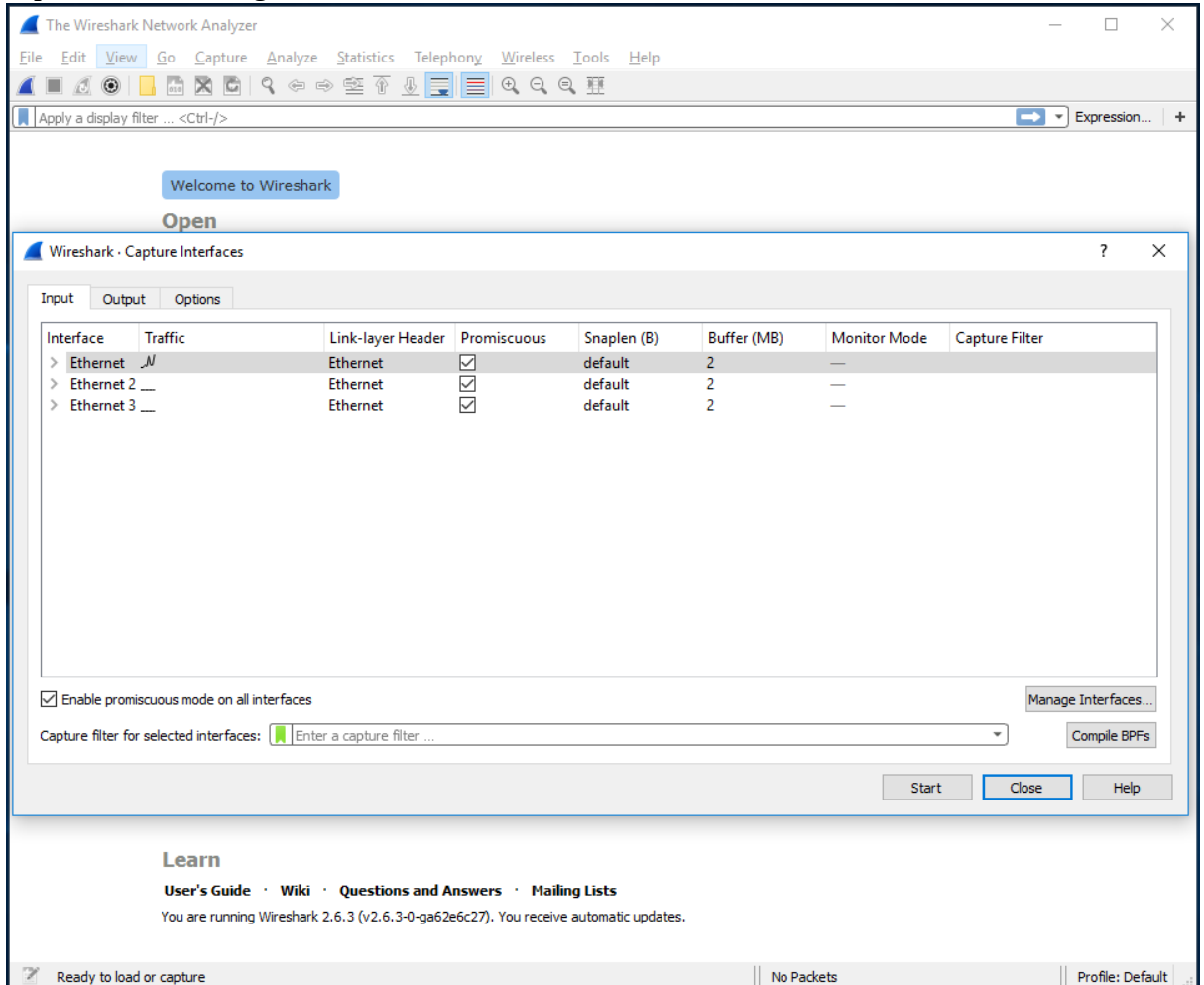


Figure A.4: Capture and Options Menu

- Once you begin packet capture, result will be shown as in Figure A.5.

The image shows the Wireshark interface with a packet capture in progress. The top pane displays a list of 90 captured packets. The bottom pane shows the details of the selected packet (No. 90), which is an Ethernet II frame from Dell_23:e3:3a to Cisco_d5:79:ff, containing an Internet Protocol Version 4 packet from 10.60.82.216 to 52.230.84.0, a Transmission Control Protocol packet with port 49724 to 443, and a Secure Sockets Layer packet.

No.	Time	Source	Destination	Protocol	Length	Info
80	4.169920	Dell_87:3b:e4	Broadcast	ARP	60	Who has 10.60.83.65? Tell 10.60.80.70
81	4.344789	10.60.82.204	255.255.255.255	DB-LSP...	255	Dropbox LAN sync Discovery Protocol
82	4.346792	10.60.82.204	255.255.255.255	DB-LSP...	255	Dropbox LAN sync Discovery Protocol
83	4.347075	10.60.82.204	10.60.83.255	DB-LSP...	255	Dropbox LAN sync Discovery Protocol
84	4.351986	HewlettP_15:15:2b	Broadcast	ARP	60	Who has 10.60.80.15? Tell 10.60.80.167
85	4.479278	Cisco_0d:7b:04	Spanning-tree-(for-...	STP	119	MST. Root = 0/10/00:23:04:ee:be:03 Cost = 20004 Port
86	4.530418	10.60.83.171	239.255.255.250	SSDP	216	M-SEARCH * HTTP/1.1
87	4.530808	10.60.83.171	239.255.255.250	SSDP	216	M-SEARCH * HTTP/1.1
88	4.700996	Dell_75:07:b3	Broadcast	ARP	60	Who has 10.60.80.162? Tell 10.60.80.64
89	4.767952	10.60.82.204	10.60.83.255	NBNS	92	Name query NB MSI-PC<1c>
90	4.995012	10.60.80.78	255.255.255.255	DB-LSP...	176	Dropbox LAN sync Discovery Protocol

Frame 1: 107 bytes on wire (856 bits), 107 bytes captured (856 bits) on interface 0
 Ethernet II, Src: Dell_23:e3:3a (00:24:e8:23:e3:3a), Dst: Cisco_d5:79:ff (00:14:6a:d5:79:ff)
 Internet Protocol Version 4, Src: 10.60.82.216, Dst: 52.230.84.0
 Transmission Control Protocol, Src Port: 49724, Dst Port: 443, Seq: 1, Ack: 1, Len: 53
 Secure Sockets Layer

0000 00 14 6a d5 79 ff 00 24 e8 23 e3 3a 08 00 45 00 ..j.y..\$ #.:..E.
 0010 00 5d 64 e9 40 00 80 06 00 00 0a 3c 52 d8 34 e6 ..]d.@...<R.4.
 0020 54 00 c2 3c 01 bb 11 3f 6d 43 3e d4 b9 24 50 18 T...<...? mC>..\$P.
 0030 01 01 e6 49 00 00 17 03 01 00 30 07 75 ee af 3c ...I.....0-u...<
 0040 42 81 47 9e 52 29 c3 be ea 2e 88 ac 28 ed bf e6 B.G.R)...((...
 0050 20 36 00 a7 ad f6 ff 8a b0 6d 63 a3 c7 a4 68 ca 6.....mc...h.
 0060 fd 6d b3 05 36 73 0e 3c 54 bc 07 ..m...6s.<T..

Figure A.5: Wireshark packet capture result

- By selecting Capture pulldown menu and selecting Stop, you can stop packet capture.

- Type “arp” in packet display filter field and press Enter key. This will cause only ARP message to be displayed in the packet-listing window as shown in Figure A.6.

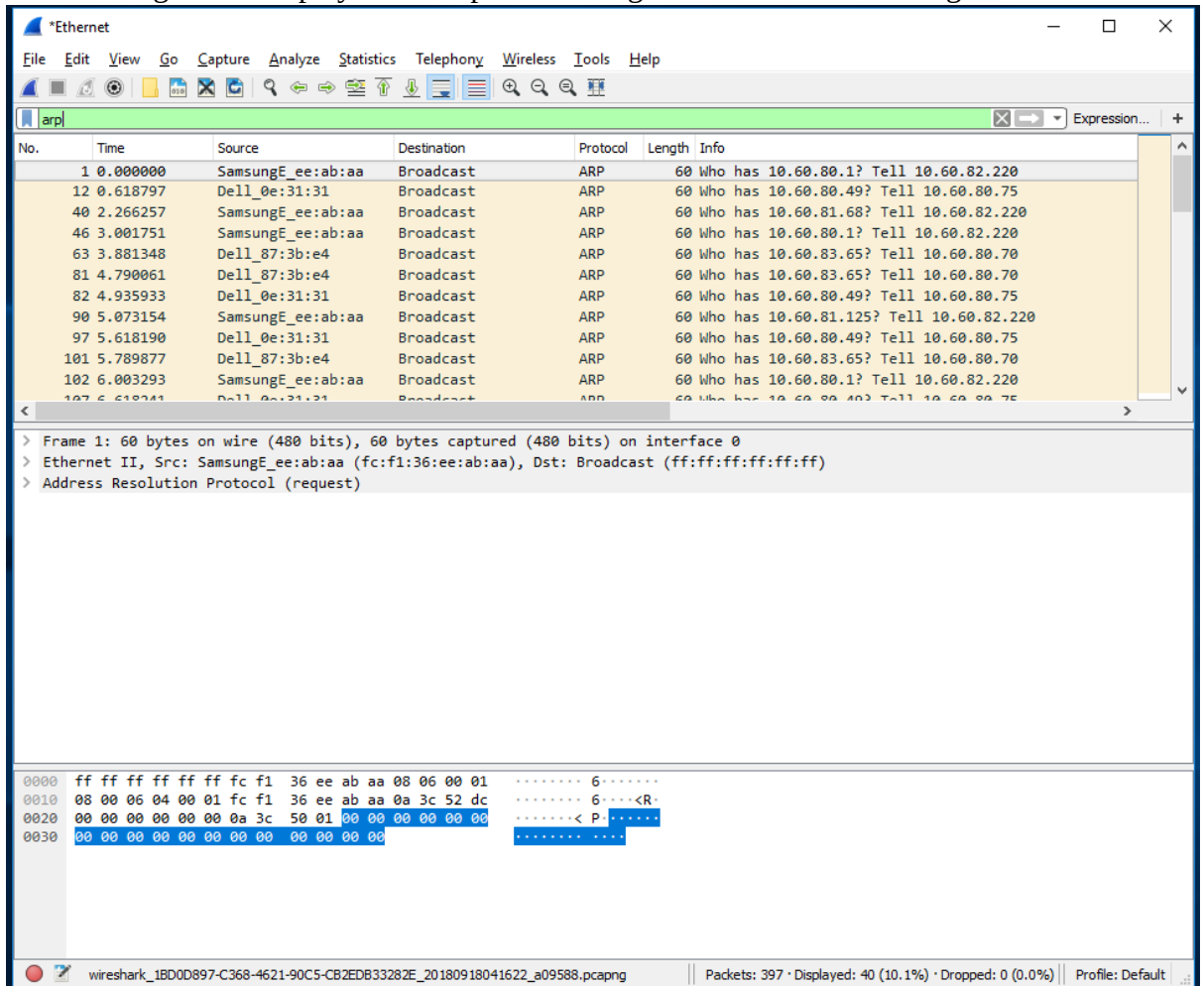


Figure A.6: ARP packet capture

- To save the trace result, use File pulldown menu and select Save function as shown in Figure A.7.

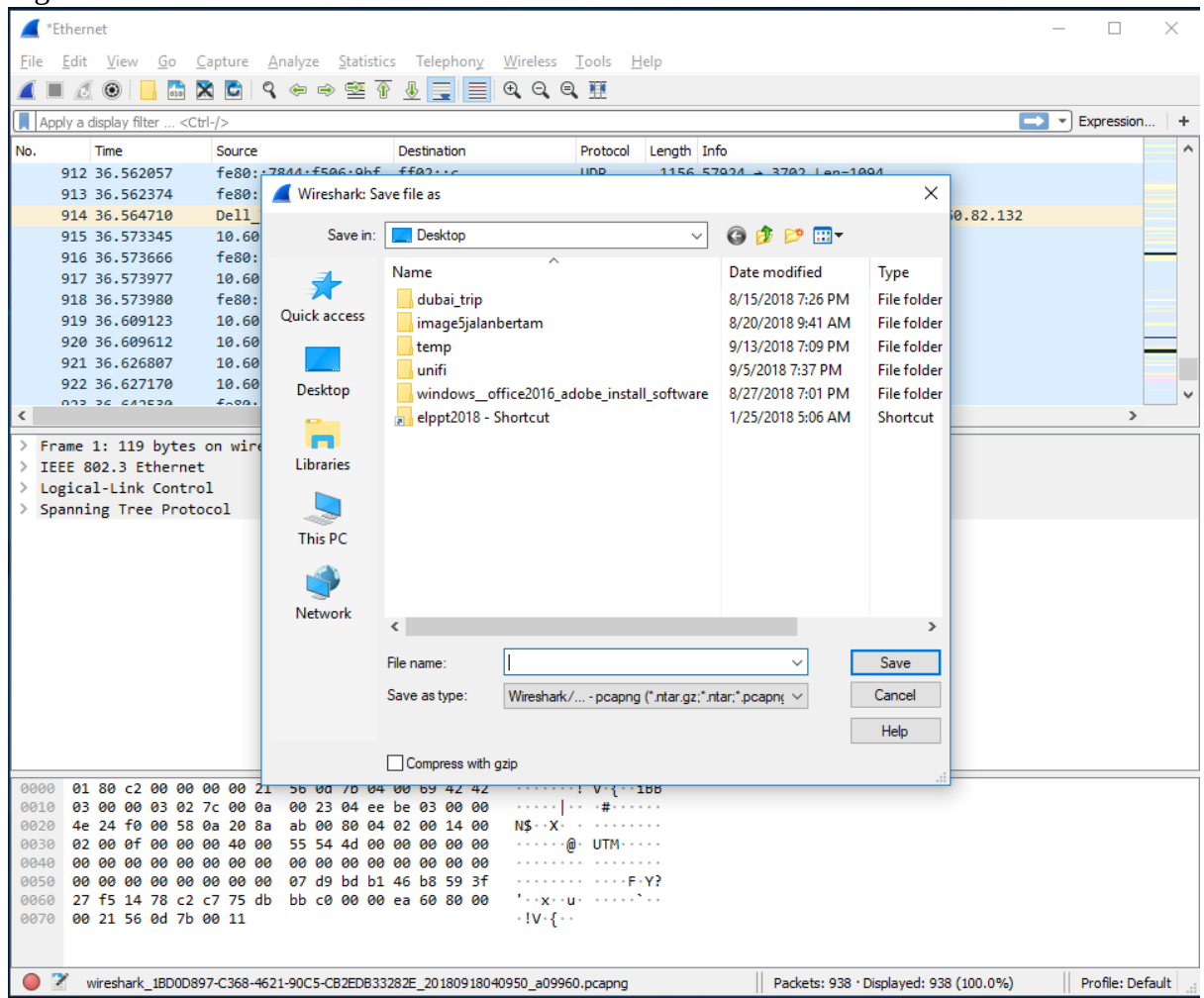


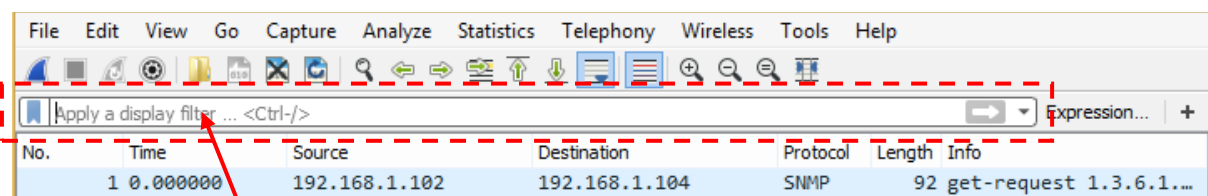
Figure A.7: Save Wireshark trace result

PART B: HTTP Trace

In this part, we'll explore several aspects of the HTTP protocol: the basic GET/response interaction, HTTP message formats and retrieving HTML files with embedded objects. Before beginning these labs, you might want to review Section 2.2 of the textbook.

B.1 The Basic HTTP GET/response interaction

- Open packet trace file **lab1-http-B01.pcapng**.
- Enter “**http**” (just the letters, not the quotation marks) in the **packet display filter field**, so that only captured HTTP messages will be displayed later in the packet-listing window. Refer to figure below:



packet display filter

- By looking at the information in the HTTP GET and response messages, answer the following questions:
 1. What version of HTTP is the server running?
[HTTP/1.1](#)
 2. What is the IP address of the client computer?
[192.168.1.102](#)
 3. What is the IP address of the gaia.cs.umass.edu server?
[128.119.245.12](#)
 4. How many bytes of content are being returned to client browser?
[439 bytes are returned for the request for /ethereal-labs/ lab2-1.html](#)
[1395 bytes are returned for the request for /favicon.ico](#)
[Total: 1834 bytes are returned.](#)
 5. What is the status code returned from the server to client browser?
[Status code 200 is returned for the request for /ethereal-labs/ lab2-1.html](#)
[Status code 404 is returned for the request for /favicon.ico](#)

B.2 The HTTP CONDITIONAL GET/response interaction

- Open packet trace file **lab1-http-B02.pcapng**.
- By looking at the information in the HTTP GET and response messages, answer the following questions:
 1. Inspect the contents of the first HTTP GET request from your browser to the server. Do you see an “IF-MODIFIED-SINCE” line in the HTTP GET?
No
 2. Inspect the contents of the server response after the first GET request from client. Did the server explicitly return the contents of the file? How can you tell?
Yes. We can see that the HTML text is sent back to the client.
 3. Now inspect the contents of the second HTTP GET request from your browser to the server. Do you see an “IF-MODIFIED-SINCE:” line in the HTTP GET? If so, what information follows the “IF-MODIFIED-SINCE:” header?
Yes. It follows by the date Tue, 23 Sep 2003 05:35:00 GMT. It tells the server to send the object only if the object has been modified since the date.
 4. What is the HTTP status code and phrase returned from the server in response to this second HTTP GET? Did the server explicitly return the contents of the file? Explain.
304 Not Modified. The server did not explicitly return the contents of the file. The status code will tell the proxy cache that it can forward the cached copy of the object to the requesting browser.

B.3 HTML Documents with Embedded Objects

- Open packet trace file **lab1-http-B03.pcapng**.
- By looking at the information in the HTTP GET and response messages, answer the following questions:

1. How many HTTP GET request messages did client browser send?

3

2. To which Internet addresses were these GET requests sent?

<http://gaia.cs.umass.edu/ethereal-labs/lab2-4.html>

<http://www.aw-bc.com/catalog/images/pearson-logo-footer.gif>

<http://manic.cs.umass.edu/~kurose/cover.jpg>

3. any bytes of content are being returned to client browser for the **pearson-logo-footer.gif** image file?

3357 bytes

4. How many bytes of content are being returned to client browser for the **cover.jpg** image file?

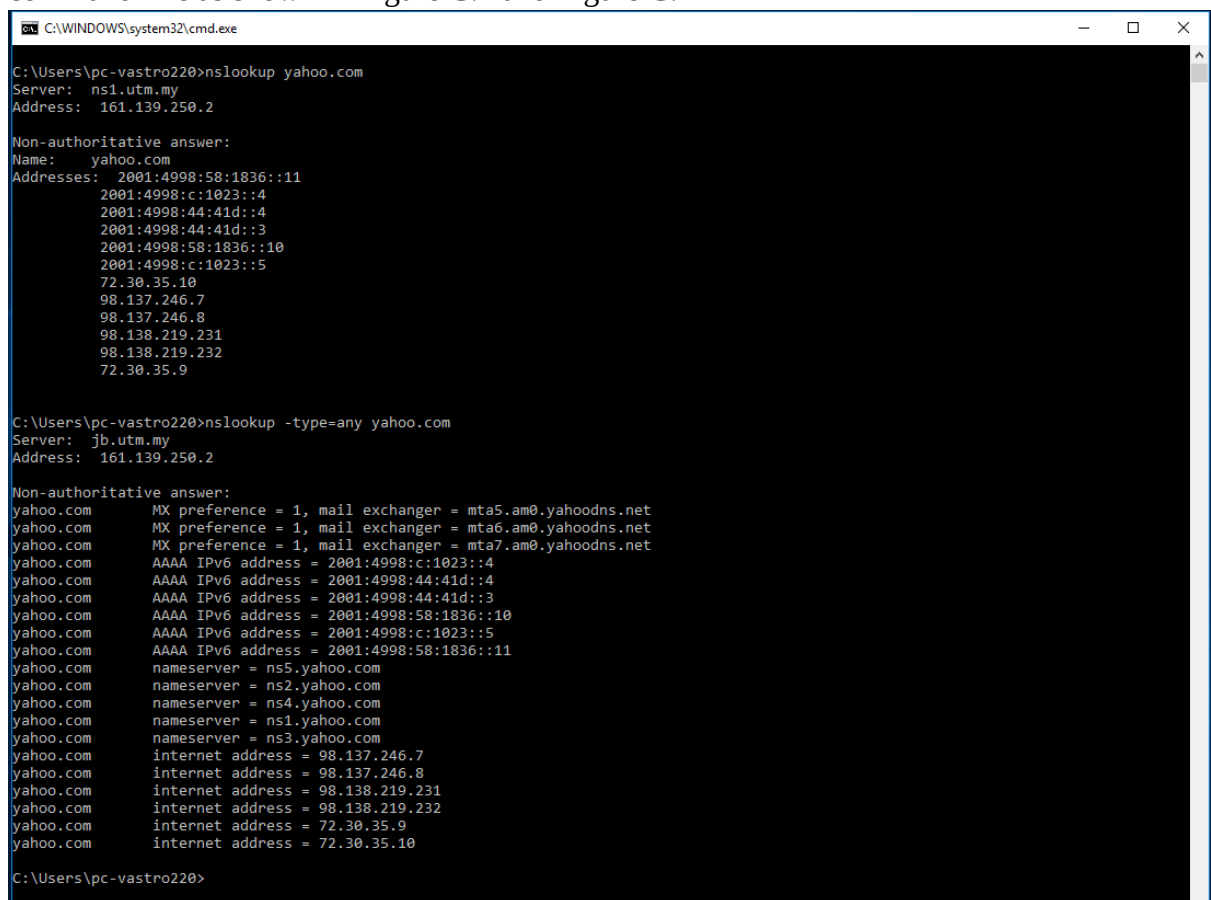
15642 bytes

PART C: DNS Trace

1.0 nslookup

nslookup tool allows the host running the tool to query any specified DNS server for a DNS record. The queried DNS server can be a root DNS server, a top-level-domain DNS server, an authoritative DNS server, or an intermediate DNS server. To accomplish this task, nslookup sends a DNS query to the specified DNS server, receives a DNS reply from that same DNS server, and displays the result.

- To run it in Windows, open the Command Prompt (cmd) and run nslookup on the command line as shown in Figure C.1 and Figure C.2



```
C:\WINDOWS\system32\cmd.exe

C:\Users\pc-vastro220>nslookup yahoo.com
Server:  ns1.utm.my
Address:  161.139.250.2

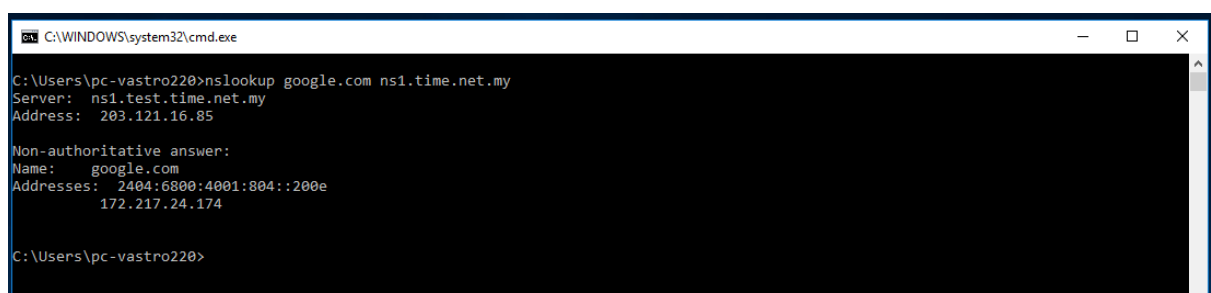
Non-authoritative answer:
Name:     yahoo.com
Addresses: 2001:4998:58:1836::11
           2001:4998:c:1023::4
           2001:4998:44:41d::4
           2001:4998:44:41d::3
           2001:4998:58:1836::10
           2001:4998:c:1023::5
           72.30.35.10
           98.137.246.7
           98.137.246.8
           98.138.219.231
           98.138.219.232
           72.30.35.9

C:\Users\pc-vastro220>nslookup -type=any yahoo.com
Server:  jb.utm.my
Address:  161.139.250.2

Non-authoritative answer:
yahoo.com      MX preference = 1, mail exchanger = mta5.am0.yahoodns.net
yahoo.com      MX preference = 1, mail exchanger = mta6.am0.yahoodns.net
yahoo.com      MX preference = 1, mail exchanger = mta7.am0.yahoodns.net
yahoo.com      AAAA IPv6 address = 2001:4998:c:1023::4
yahoo.com      AAAA IPv6 address = 2001:4998:44:41d::4
yahoo.com      AAAA IPv6 address = 2001:4998:44:41d::3
yahoo.com      AAAA IPv6 address = 2001:4998:58:1836::10
yahoo.com      AAAA IPv6 address = 2001:4998:c:1023::5
yahoo.com      AAAA IPv6 address = 2001:4998:58:1836::11
yahoo.com      nameserver = ns5.yahoo.com
yahoo.com      nameserver = ns2.yahoo.com
yahoo.com      nameserver = ns4.yahoo.com
yahoo.com      nameserver = ns1.yahoo.com
yahoo.com      nameserver = ns3.yahoo.com
yahoo.com      internet address = 98.137.246.7
yahoo.com      internet address = 98.137.246.8
yahoo.com      internet address = 98.138.219.231
yahoo.com      internet address = 98.138.219.232
yahoo.com      internet address = 72.30.35.9
yahoo.com      internet address = 72.30.35.10

C:\Users\pc-vastro220>
```

Figure C.1: nslookup result



```
C:\WINDOWS\system32\cmd.exe

C:\Users\pc-vastro220>nslookup google.com ns1.time.net.my
Server:  ns1.test.time.net.my
Address:  203.121.16.85

Non-authoritative answer:
Name:     google.com
Addresses: 2404:6800:4001:804::200e
           172.217.24.174

C:\Users\pc-vastro220>
```

Figure C.2: nslookup result

1. Run nslookup to obtain the IP address of a www.microsoft.com server. What is the IP address of that server? Add screenshot to your answer.

```
C:\Users\yuyla>nslookup microsoft.com
Server: ns3.utm.my
Address: 161.139.168.168

Non-authoritative answer:
Name: microsoft.com
Addresses: 2603:1010:3:3::5b
           2603:1030:c02:8::14
           2603:1020:201:10::10f
           2603:1030:b:3::152
           2603:1030:20e:3::23c
           20.76.201.171
           20.231.239.246
           20.70.246.20
           20.112.250.133
           20.236.44.162

C:\Users\yuyla>nslookup -type=any microsoft.com
Server: ns3.utm.my
Address: 161.139.168.168

Non-authoritative answer:
microsoft.com MX preference = 10, mail exchanger = microsoft-com.mail.protection.outlook.com
microsoft.com AAAA IPv6 address = 2603:1030:b:3::152
microsoft.com AAAA IPv6 address = 2603:1030:c02:8::14
microsoft.com AAAA IPv6 address = 2603:1020:201:10::10f
microsoft.com AAAA IPv6 address = 2603:1010:3:3::5b
microsoft.com AAAA IPv6 address = 2603:1030:20e:3::23c
microsoft.com internet address = 20.236.44.162
microsoft.com internet address = 20.70.246.20
microsoft.com internet address = 20.112.250.133
microsoft.com internet address = 20.76.201.171
microsoft.com internet address = 20.231.239.246
microsoft.com nameserver = ns3-39.azure-dns.org
microsoft.com nameserver = ns4-39.azure-dns.info
microsoft.com nameserver = ns1-39.azure-dns.com
microsoft.com nameserver = ns2-39.azure-dns.net
microsoft.com HINFO CPU = RFC8482
```

2. Run nslookup to determine the non-authoritative DNS servers for domain microsoft.com. Add screenshot to your answer.

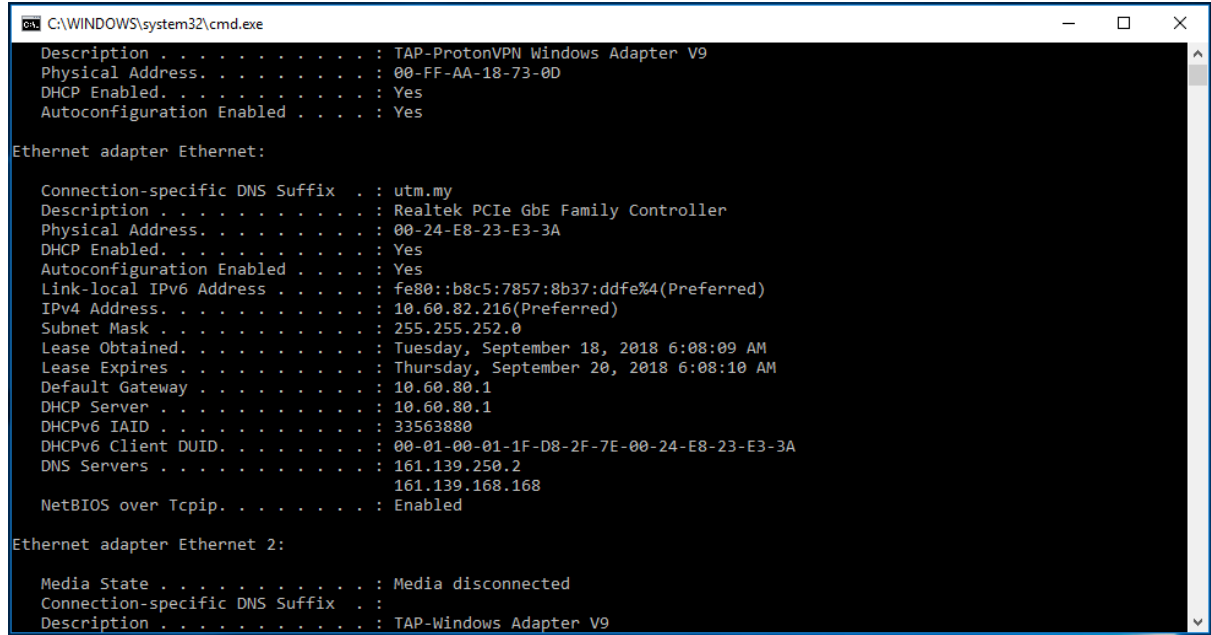
```
C:\Users\yuyla>nslookup microsoft.com ns1.time.net.my
Server: ns1.test.time.net.my
Address: 203.121.16.85

Non-authoritative answer:
Name: microsoft.com
Addresses: 2603:1030:20e:3::23c
           2603:1010:3:3::5b
           2603:1020:201:10::10f
           2603:1030:b:3::152
           2603:1030:c02:8::14
           20.236.44.162
           20.76.201.171
           20.112.250.133
           20.70.246.20
           20.231.239.246
```

2.0 ipconfig

ipconfig can be used to show your current TCP/IP information, including your address, DNS server addresses, adapter type and so on.

- Information about host, use the following command: ipconfig /all



```
C:\WINDOWS\system32\cmd.exe
Description . . . . . : TAP-ProtonVPN Windows Adapter V9
Physical Address. . . . . : 00-FF-AA-18-73-0D
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes

Ethernet adapter Ethernet:

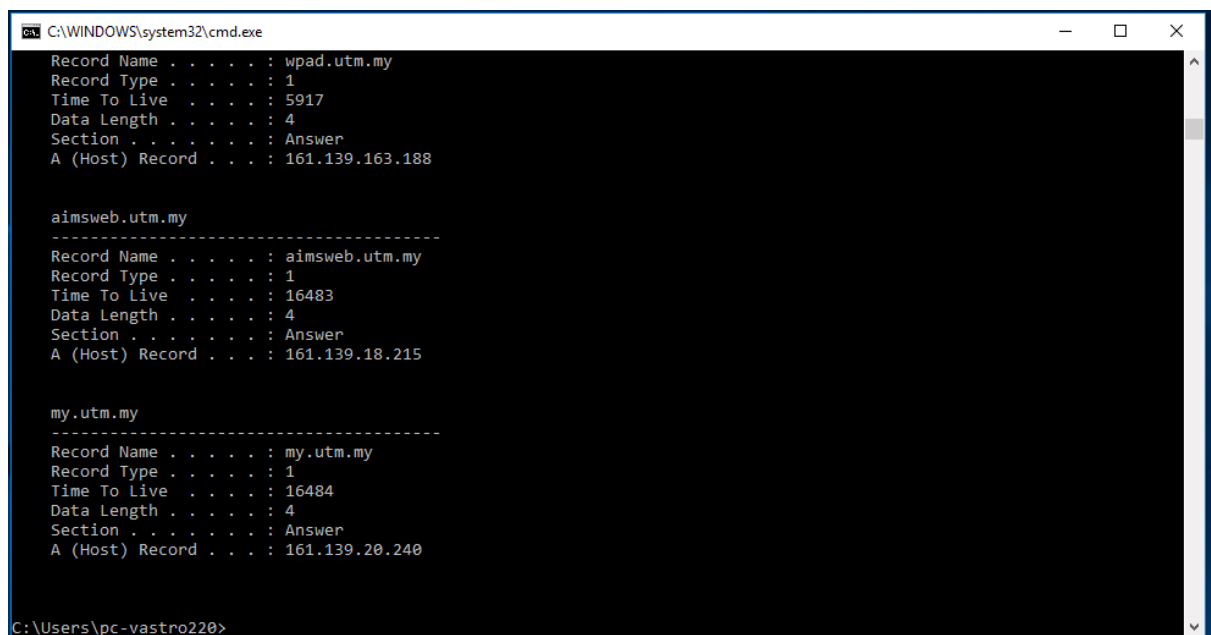
    Connection-specific DNS Suffix  . : utm.my
    Description . . . . . : Realtek PCIe GbE Family Controller
    Physical Address. . . . . : 00-24-E8-23-E3-3A
    DHCP Enabled. . . . . : Yes
    Autoconfiguration Enabled . . . . : Yes
    Link-local IPv6 Address . . . . . : fe80::b8c5:7857:8b37:ddfe%4(Preferred)
    IPv4 Address. . . . . : 10.60.82.216(Preferred)
    Subnet Mask . . . . . : 255.255.252.0
    Lease Obtained. . . . . : Tuesday, September 18, 2018 6:08:09 AM
    Lease Expires . . . . . : Thursday, September 20, 2018 6:08:10 AM
    Default Gateway . . . . . : 10.60.80.1
    DHCP Server . . . . . : 10.60.80.1
    DHCPv6 IAID . . . . . : 33563880
    DHCPv6 Client DUID. . . . . : 00-01-00-01-1F-D8-2F-7E-00-24-E8-23-E3-3A
    DNS Servers . . . . . : 161.139.250.2
                           161.139.168.168
    NetBIOS over Tcpip. . . . . : Enabled

Ethernet adapter Ethernet 2:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :
    Description . . . . . : TAP-Windows Adapter V9
```

Figure C.3: ipconfig /all result

- ipconfig is also very useful for managing the DNS information stored in your host. Each entry shows the remaining Time to Live (TTL) in seconds.
Command: ipconfig /displaydns



```
C:\WINDOWS\system32\cmd.exe
Record Name . . . . . : wpad.utm.my
Record Type . . . . . : 1
Time To Live . . . . . : 5917
Data Length . . . . . : 4
Section . . . . . : Answer
A (Host) Record . . . : 161.139.163.188

-----
aimsweb.utm.my
Record Name . . . . . : aimsweb.utm.my
Record Type . . . . . : 1
Time To Live . . . . . : 16483
Data Length . . . . . : 4
Section . . . . . : Answer
A (Host) Record . . . : 161.139.18.215

-----
my.utm.my
Record Name . . . . . : my.utm.my
Record Type . . . . . : 1
Time To Live . . . . . : 16484
Data Length . . . . . : 4
Section . . . . . : Answer
A (Host) Record . . . : 161.139.20.240

C:\Users\pc-vastro220>
```

Figure C.4: ipconfig /displaydns result

- Flushing the DNS cache clears all entries and reloads the entries from the hosts file.

Command: ipconfig /flushdns



```
C:\WINDOWS\system32\cmd.exe
C:\Users\pc-vastro220>ipconfig /flushdns

Windows IP Configuration

Successfully flushed the DNS Resolver Cache.

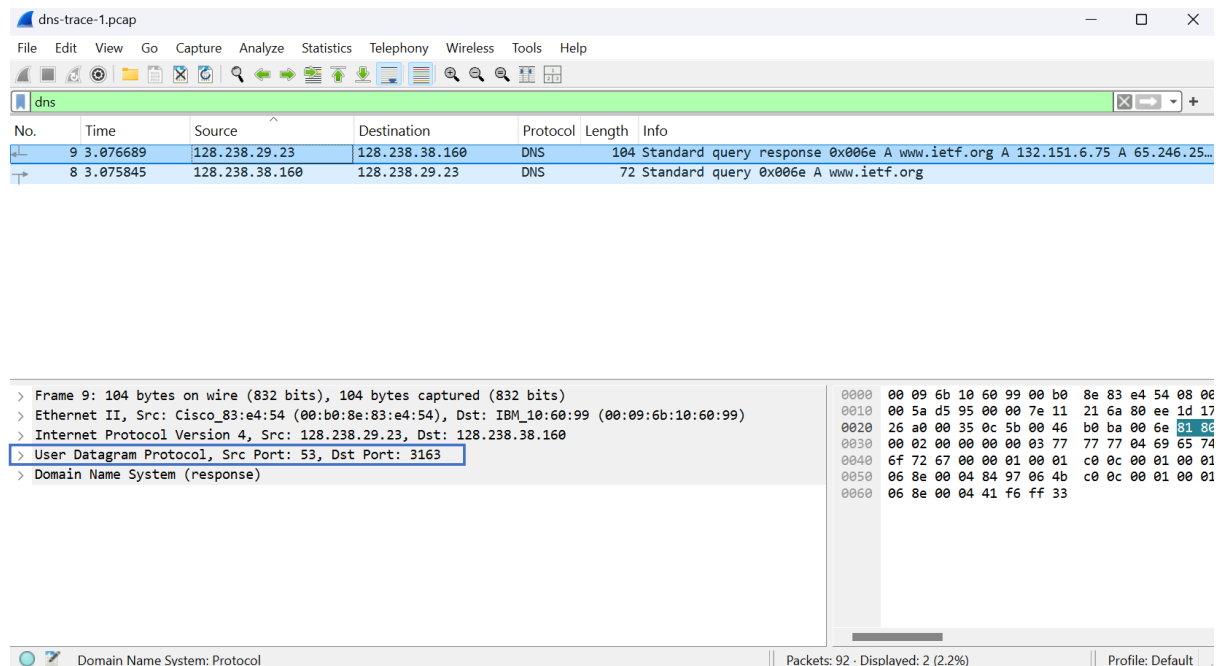
C:\Users\pc-vastro220>
```

Figure C.5: ipconfig /flushdns result

3.0 Tracing DNS with Wireshark

- Open packet trace file dns-trace-1. Answer the following questions.
- 1. Locate the DNS query and response messages. Are then sent over UDP or TCP? Add screenshots in your answer.

UDP



dns-trace-1.pcap

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

No.	Time	Source	Destination	Protocol	Length	Info
9	3.076689	128.238.29.23	128.238.38.160	DNS	104	Standard query response 0x006e A www.ietf.org A 132.151.6.75 A 65.246.25...
8	3.075845	128.238.38.160	128.238.29.23	DNS	72	Standard query 0x006e A www.ietf.org

> Frame 9: 104 bytes on wire (832 bits), 104 bytes captured (832 bits)
> Ethernet II, Src: Cisco_83:e4:54 (00:b0:8e:83:e4:54), Dst: IBM_10:60:99 (00:09:6b:10:60:99)
> Internet Protocol Version 4, Src: 128.238.29.23, Dst: 128.238.38.160
> User Datagram Protocol, Src Port: 53, Dst Port: 3163
> Domain Name System (response)

0000 00 09 6b 10 60 99 00 b0 8e 83 e4 54 08 0e
0010 00 5a d5 95 00 00 7e 11 21 6a 80 ee 1d 17
0020 26 a0 00 35 0c 5b 00 46 b0 ba 00 6e 81 80
0030 00 02 00 00 00 00 03 77 77 77 04 69 65 74
0040 6f 72 67 00 00 01 00 01 c0 0c 00 01 00 01
0050 06 8e 00 04 84 97 06 4b c0 0c 00 01 00 01
0060 06 8e 00 04 41 f6 ff 33

Domain Name System: Protocol | Packets: 92 · Displayed: 2 (2.2%) | Profile: Default

2. What is the destination port for the DNS query message? What is the source port of DNS response message? Add screenshots in your answer.

Destination Port: 53

Source Port: 3163

The screenshot shows a Wireshark capture of a DNS query and response. The packet list shows two packets: a DNS query (No. 8) and a DNS response (No. 9). The query is from 128.238.38.160 to 128.238.29.23. The response is from 128.238.29.23 to 128.238.38.160. The packet details for packet 8 show the User Datagram Protocol (UDP) source port as 3163 and destination port as 53. The packet bytes show the DNS query structure.

No.	Time	Source	Destination	Protocol	Length	Info
9	3.076689	128.238.29.23	128.238.38.160	DNS	104	Standard query response 0x006e A www.ietf.org A 132.151.6.75 A 65.246.25...
8	3.075845	128.238.38.160	128.238.29.23	DNS	72	Standard query 0x006e A www.ietf.org

Frame 8: 72 bytes on wire (576 bits), 72 bytes captured (576 bits)
> Ethernet II, Src: IBM_10:60:99 (00:09:6b:10:60:99), Dst: All-HSRP-routers_00 (00:00:0c:07:ac:00)
> Internet Protocol Version 4, Src: 128.238.38.160, Dst: 128.238.29.23
> User Datagram Protocol, Src Port: 3163, Dst Port: 53
> Domain Name System (query)

0000 00 00 0c 07 ac 00 00 09 6b 10 60 99 08 00
0010 00 3a 22 9e 00 00 00 d2 81 80 ee 26 a0
0020 1d 17 0c 5b 00 35 00 26 8a cb 00 6e 01 00
0030 00 00 00 00 00 03 77 77 77 04 69 65 74
0040 6f 72 67 00 00 01 00 01

3. To what IP address is the DNS query message sent? Add screenshots in your answer.

128.238.29.23

The screenshot shows a Wireshark capture of a DNS query and response. The packet list shows two packets: a DNS query (No. 8) and a DNS response (No. 9). The query is from 128.238.38.160 to 128.238.29.23. The response is from 128.238.29.23 to 128.238.38.160. The packet details for packet 8 show the User Datagram Protocol (UDP) source port as 3163 and destination port as 53. The packet bytes show the DNS query structure.

No.	Time	Source	Destination	Protocol	Length	Info
9	3.076689	128.238.29.23	128.238.38.160	DNS	104	Standard query response 0x006e A www.ietf.org A 132.151.6.75 A 65.246.25...
8	3.075845	128.238.38.160	128.238.29.23	DNS	72	Standard query 0x006e A www.ietf.org

Frame 8: 72 bytes on wire (576 bits), 72 bytes captured (576 bits)
> Ethernet II, Src: IBM_10:60:99 (00:09:6b:10:60:99), Dst: All-HSRP-routers_00 (00:00:0c:07:ac:00)
> Internet Protocol Version 4, Src: 128.238.38.160, Dst: 128.238.29.23
> User Datagram Protocol, Src Port: 3163, Dst Port: 53
> Domain Name System (query)

0000 00 00 0c 07 ac 00 00 09 6b 10 60 99 08 00
0010 00 3a 22 9e 00 00 00 d2 81 80 ee 26 a0
0020 1d 17 0c 5b 00 35 00 26 8a cb 00 6e 01 00
0030 00 00 00 00 00 03 77 77 77 04 69 65 74
0040 6f 72 67 00 00 01 00 01

- Examine the DNS query message. What “Type” of DNS query is it? Does the query message contain any “answers”? Add screenshots in your answer.

Type A query.

The query does not contain any “answer”.

The screenshot shows a Wireshark capture of a DNS query message. The packet list at the top shows two packets: a DNS query (No. 9, Time 3.076689, Source 128.238.29.23, Destination 128.238.38.160, Protocol DNS, Length 104) and a DNS response (No. 8, Time 3.075845, Source 128.238.38.160, Destination 128.238.29.23, Protocol DNS, Length 72). The packet details pane shows the selected packet (No. 9) as a 'Domain Name System (query)' with Transaction ID 0x006e, Flags 0x0100, and one question: 'www.ietf.org: type A, class IN'. The packet bytes pane shows the raw data of the query.

- Examine the DNS response message. How many “answers” are provided? What do each of these answers contain? Add screenshots in your answer.

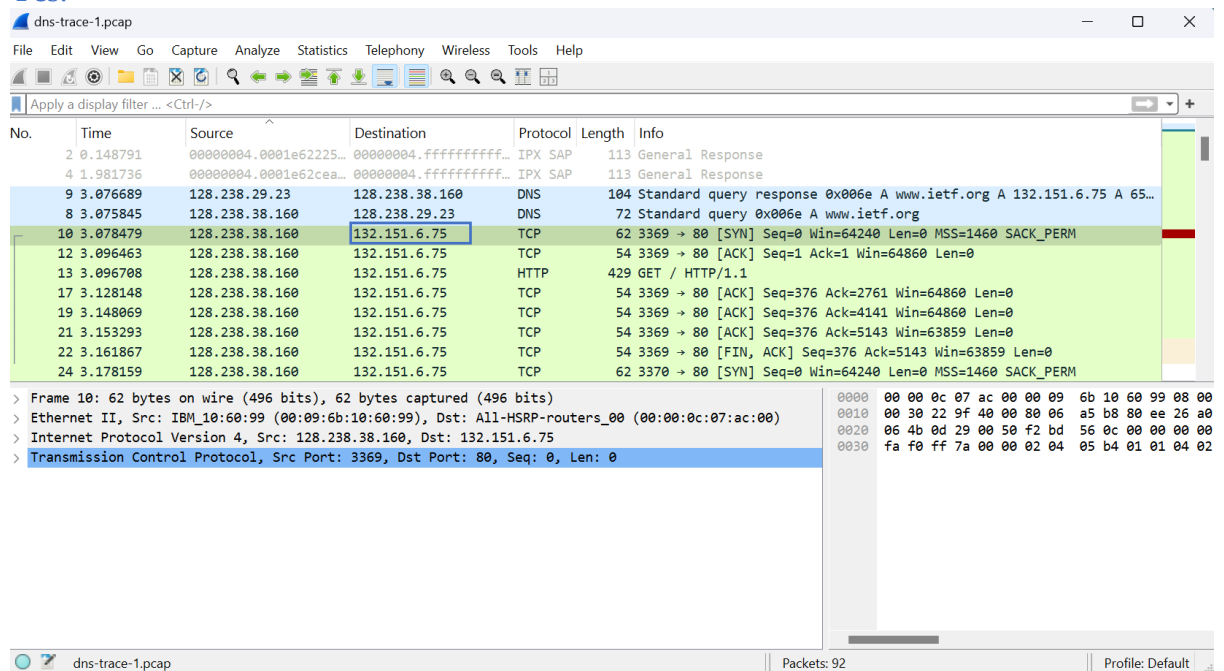
2 “answers” are provided

Each of the answer gives the IP address of the hostname queried.

The screenshot shows a Wireshark capture of a DNS response message. The packet list at the top shows two packets: a DNS query (No. 9, Time 3.076689, Source 128.238.29.23, Destination 128.238.38.160, Protocol DNS, Length 104) and a DNS response (No. 8, Time 3.075845, Source 128.238.38.160, Destination 128.238.29.23, Protocol DNS, Length 72). The packet details pane shows the selected packet (No. 8) as a 'Domain Name System (response)' with Transaction ID 0x006e, Flags 0x8180, and two answers: 'www.ietf.org: type A, class IN, addr 132.151.6.75' and 'www.ietf.org: type A, class IN, addr 65.246.255.51'. The packet bytes pane shows the raw data of the response.

6. Consider the subsequent TCP SYN packet sent by your host. Does the destination IP address of the SYN packet correspond to any of the IP addresses provided in the DNS response message?

Yes.



No.	Time	Source	Destination	Protocol	Length	Info
2	0.148791	00000004.0001e62225...	00000004.ffffffffffff...	IPX SAP	113	General Response
4	1.981736	00000004.0001e62cea...	00000004.ffffffffffff...	IPX SAP	113	General Response
9	3.076689	128.238.29.23	128.238.38.160	DNS	104	Standard query response 0x006e A www.ietf.org A 132.151.6.75 A 65...
8	3.075845	128.238.38.160	128.238.29.23	DNS	72	Standard query 0x006e A www.ietf.org
10	3.078479	128.238.38.160	132.151.6.75	TCP	62	3369 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM
12	3.096463	128.238.38.160	132.151.6.75	TCP	54	3369 → 80 [ACK] Seq=1 Ack=1 Win=64860 Len=0
13	3.096708	128.238.38.160	132.151.6.75	HTTP	429	GET / HTTP/1.1
17	3.128148	128.238.38.160	132.151.6.75	TCP	54	3369 → 80 [ACK] Seq=376 Ack=2761 Win=64860 Len=0
19	3.148069	128.238.38.160	132.151.6.75	TCP	54	3369 → 80 [ACK] Seq=376 Ack=4141 Win=64860 Len=0
21	3.153293	128.238.38.160	132.151.6.75	TCP	54	3369 → 80 [ACK] Seq=376 Ack=5143 Win=63859 Len=0
22	3.161867	128.238.38.160	132.151.6.75	TCP	54	3369 → 80 [FIN, ACK] Seq=376 Ack=5143 Win=63859 Len=0
24	3.178159	128.238.38.160	132.151.6.75	TCP	62	3370 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM

> Frame 10: 62 bytes on wire (496 bits), 62 bytes captured (496 bits)
 > Ethernet II, Src: IBM_10:60:99 (00:09:6b:10:60:99), Dst: All-HSRP-routers_00 (00:00:0c:07:ac:00)
 > Internet Protocol Version 4, Src: 128.238.38.160, Dst: 132.151.6.75
 > Transmission Control Protocol, Src Port: 3369, Dst Port: 80, Seq: 0, Len: 0

7. This web page contains images. Before retrieving each image, does your host issue new DNS queries?

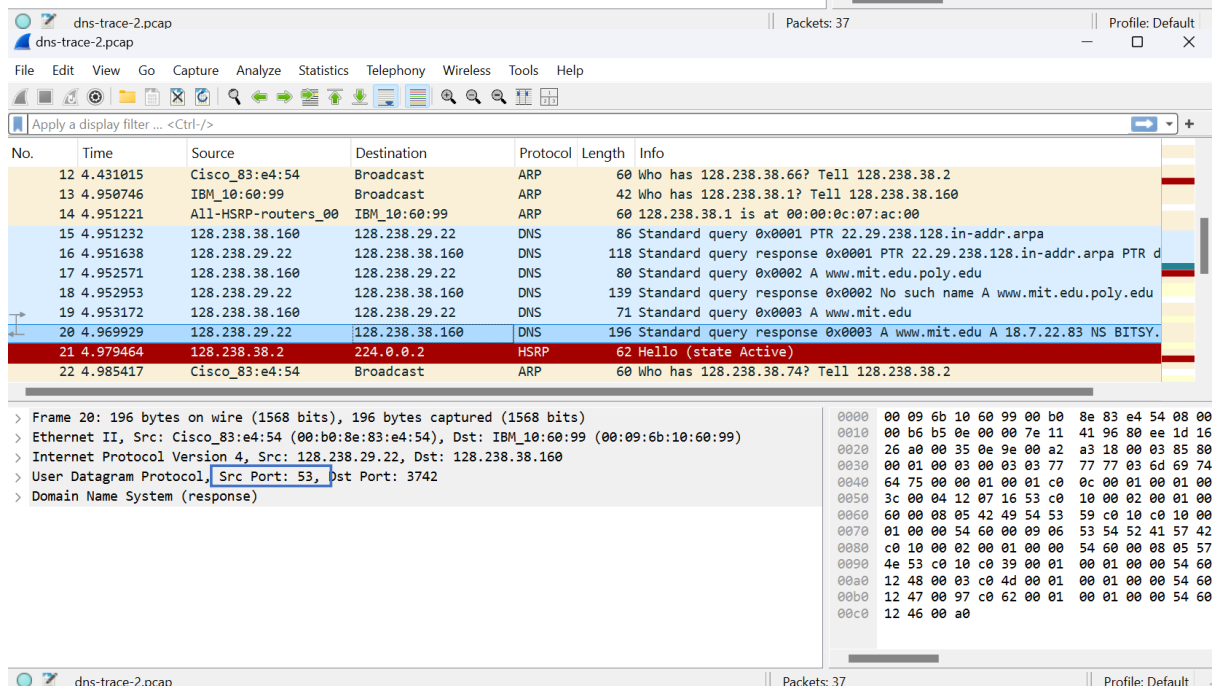
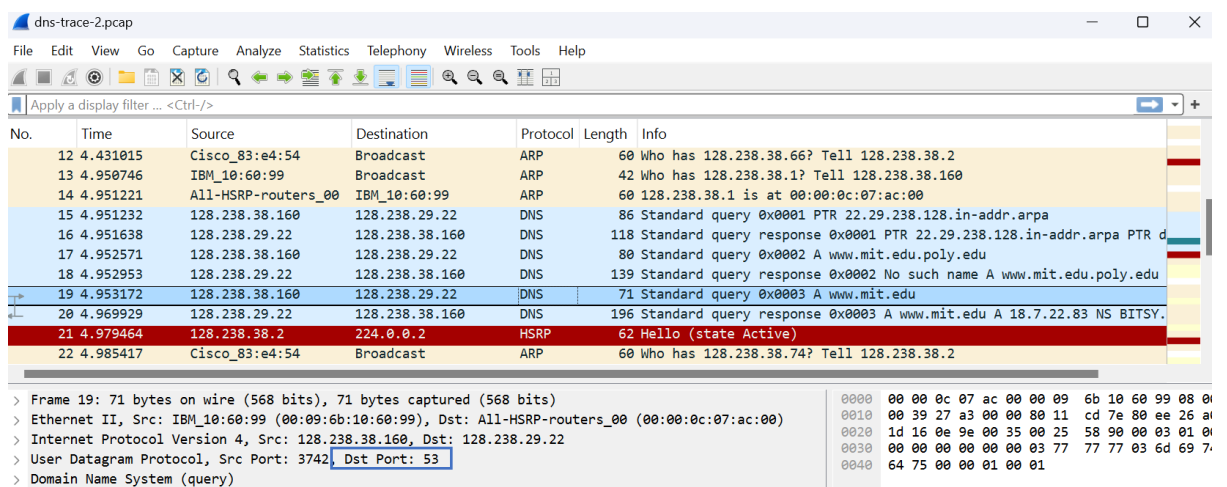
No

- Open packet trace file dns-trace-2 for nslookup.
- We see from Wireshark that nslookup actually sent three DNS queries and received three DNS responses. For the purpose of this lab, ignore the first two sets of queries/responses, as they are specific to nslookup and are not normally generated by standard Internet applications. You should instead focus on the last query and response messages.
- Answer the following questions.

8. What is the destination port for the DNS query message? What is the source port of DNS response message? Add screenshots in your answer.

Destination port for DNS query message: 53

Source port for DNS response message: 53



9. To what IP address is the DNS query message sent? Is this the IP address of your default local DNS server? Add screenshots in your answer.

Query Destination IP: 128.238.29.22

Default Local DNS IP: 161.139.168.168

The IP address are not the same.

The screenshot shows a Wireshark packet capture of a network traffic. The packet list on the left shows a series of packets, including ARP requests and DNS queries. Packet 19 is highlighted, showing a DNS query from 128.238.38.160 to 128.238.29.22. The packet details pane on the right shows the structure of the DNS query, including the query ID, flags, and the query name. The packet bytes pane on the right shows the raw data of the packet.

```
Internet Protocol Version 4 (ip), 20 bytes
Packets: 37
Profile: Default

C:\Users\yuyila>ipconfig /all

Windows IP Configuration

Host Name . . . . . : DESKTOP-3PR12MH
Primary Dns Suffix . . . . . :
Node Type . . . . . : Hybrid
IP Routing Enabled. . . . . : No
WINS Proxy Enabled. . . . . : No

Wireless LAN adapter Local Area Connection* 3:

Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . :
Description . . . . . : Microsoft Wi-Fi Direct Virtual Adapter #3
Physical Address. . . . . : 14-18-C3-B5-39-8A
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes

Wireless LAN adapter Local Area Connection* 4:

Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . :
Description . . . . . : Microsoft Wi-Fi Direct Virtual Adapter #4
Physical Address. . . . . : 16-18-C3-B5-39-89
DHCP Enabled. . . . . : No
Autoconfiguration Enabled . . . . : Yes

Wireless LAN adapter Wi-Fi:

Connection-specific DNS Suffix . :
Description . . . . . : Intel(R) Wi-Fi 6 AX201 160MHz
Physical Address. . . . . : 14-18-C3-B5-39-89
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes
Link-local IPv6 Address . . . . . : fe80::9f3d:9151:4563:bcc%12(Preferred)
IPv4 Address. . . . . : 10.205.100.63(Preferred)
Subnet Mask . . . . . : 255.255.240.0
Lease Obtained. . . . . : 17 January 2025 20:56:25
Lease Expires . . . . . : 18 January 2025 05:12:32
Default Gateway . . . . . : 10.205.101.250
DHCP Server . . . . . : 161.139.100.238
DHCPv6 IAID . . . . . : 101980355
DHCPv6 Client DUID. . . . . : 00-01-00-01-2B-12-93-9F-14-18-C3-B5-39-89
DNS Servers . . . . . : 161.139.168.168
NetBIOS over Tcpip. . . . . : Enabled
```

10. Examine the DNS query message. What “Type” of DNS query is it? Does the query message contain any “answers”? Add screenshots in your answer.

Type A query

It does not contain any “answer”

The screenshot shows a Wireshark capture of a DNS query. The packet list pane shows packet 19 as a DNS Standard query (type A, class IN) for www.mit.edu. The packet details pane shows the query structure: Transaction ID: 0x0003, Flags: 0x0100 Standard query, Questions: 1, Answer RRs: 0, Authority RRs: 0, Additional RRs: 0. The packet bytes pane shows the raw data of the query.

11. Examine the DNS response message. How many “answers” are provided? What do each of these answers contain? Add screenshots in your answer.

1 answer is provided

The answer contain the IP address of the host.

The screenshot shows a Wireshark capture of a DNS response. The packet list pane shows packet 21 as a DNS Standard query response (type A, class IN) for www.mit.edu. The packet details pane shows the response structure: Transaction ID: 0x0003, Flags: 0x8500 Standard query response, No error, Questions: 1, Answer RRs: 1, Authority RRs: 3, Additional RRs: 3. The packet bytes pane shows the raw data of the response, including the answer section with the IP address 18.7.22.83.